MIL-I-7075A
1 June 1965
Superseding
MIL-I-7075
8 January 1951

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -50 TO +50° C, 2 INCH SIZE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for single pointer, electrical resistance type temperature indicators.

1.2 Classification: The indicators shall be classified as follows:

MS28008-1	Fluorescent Dial Markings
MS28008-2	Lusterless White Dial Markings

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Method of
MIL-I-7070	Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-B-27497	Bearings, Jewel, Sapphire, or Ruby, Synthetic
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 6685

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-831	Test Reports, Preparation of
MS28008	Indicator-Temperature, Multifunction, Electrical Resistance, Single, Clamp Mounted, -50 to +50° C, 2-inch Size
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instruments
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 General. The requirements of MIL-I-7070 are applicable as requirements of this specification with the exceptions and additions specified herein. In the event of conflict, this specification shall govern.

3.3 Design and construction.

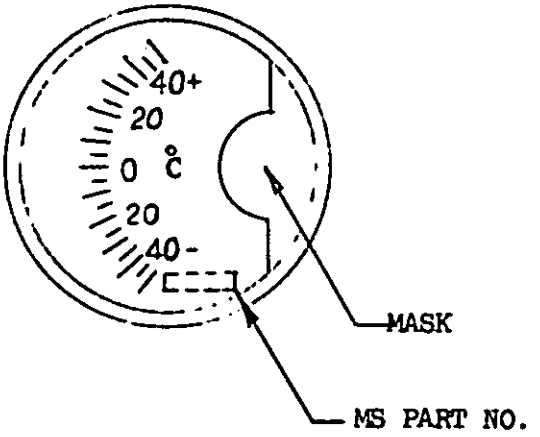
3.3.1 Case. The indicator case shall conform to MS28008.

3.3.2 Dial. The dial shall conform to Figure 1.

3.3.2.1 Dial markings. The dial markings shall be in accordance with MS33558 and Figure 1. Dimension of markings shall be as shown in the table in Figure 1. The lusterless black shall be Color 37038 of FED-STD-595.

3.3.2.2 Scale length. The scale of the indicator shall be as long as practicable and shall not be less than 1 3/4 inches as measured on the outer arc of the shortest graduation.

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MARKING	HEIGHT OR LENGTH ±0.005	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH
ALL NUMERALS	0.125	0.020	FLUORESCENT
MAJOR GRADUATIONS	0.156	0.031	LUMINESCENT
MINOR GRADUATIONS	0.093	0.020	MATERIAL
LETTERING "°C"	0.125	0.020	OR
SHADED PORTION OF POINTER	-----	-----	LUSTERLESS WHITE
BACKGROUND OF DIAL (MS PART NO.)	----- 0.060	----- 0.010	LUSTERLESS BLACK

DIMENSIONS IN INCHES.

FIGURE 1. Dial, Pointer, and Markings

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3.3.3 Pointer. The pointer shall conform to MS33585-8.

3.3.3.1 Pointer overswing. The pointer overswing shall not be less than 3 percent of full scale angular deflection beyond each end of the scale.

3.3.4 Weight. The weight of the complete indicator shall not exceed 1.4 pounds.

3.3.5 Electrical receptacle. An electrical receptacle shall be provided and located as shown on MS28008. The receptacle shall be wired as follows:

TERMINAL	CONNECTION
A	+ 24V Battery
B	Resistance Bulb
C	Common - Negative 24V Battery and remaining bulb connection
D	Open

3.3.6 Bearings, synthetic. When synthetic bearings are used they shall conform to MIL-B-27497.

3.4 Identification decalcomania. Each indicator shall be provided with 1 each of the 3 identification decalcomanias shown on Figure 2. The lettering shall conform to MS33558 and dimensions shall be as shown in the table in Figure 2. The decalcomanias shall be of the water-applied type and shall be suitable for application to the cover glass of the indicator. The decalcomanias, when applied to the cover glass of the indicator, shall be capable of meeting the tests specified in Section 4 of this specification without peeling, blistering, fading, loss of adhesion, checking, softening, or hardening.

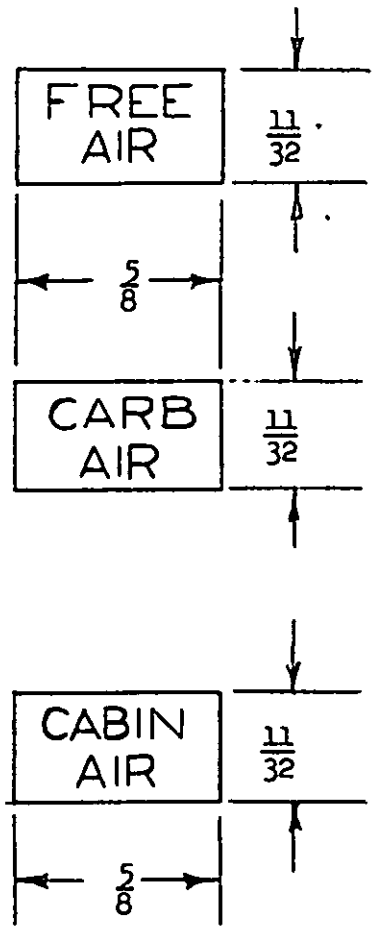
3.4.1 Each set of decalcomanias shall be packaged in an envelope marked: "THIS ENVELOPE CONTAINS IDENTIFICATION DECALCOMANIAS". Complete and detailed instructions for mounting the decalcomanias on the cover glass of the indicator shall be furnished with each set of decalcomanias.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing requirements of MIL-D-70327 shall govern changes in the manufacturer's part number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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MARKINGS	HEIGHT ± .010	WIDTH ± .005	MATERIAL OR FINISH
ALL LETTERS	0.125	.020	FLUORESCENT LUMINESCENT MATERIAL OR LUSTERLESS WHITE
BACKGROUND	-----	-----	TRANSPARENT OR LUSTERLESS BLACK

DIMENSIONS IN INCHES.

UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/32.

FIGURE 2. Decal Size and Marking

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4.2 General. Sampling, inspection and test procedures shall be in accordance with MIL-I-7070, except as specified herein.

4.3 Test Report. When tests are conducted at a location other than the laboratory of the procuring activity, three (3) copies of a test report in accordance with MIL-STD-831, shall be furnished to that activity.

4.4 Test voltage. The test voltage shall be 28.5 ± 0.5 volts DC, unless otherwise specified.

4.4.1 The test voltage for the reduced voltage operation test shall be 22.5 V DC.

4.5 Test tolerances.

4.5.1 The tolerances specified in Table I, not enclosed in parenthesis, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parenthesis, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.5.2 Scale error at room temperature. Scale errors at room temperature shall not exceed the tolerances listed in Table I.

4.5.3 Position error. Position error of the indicator shall not exceed 2° C.

4.5.4 Friction error. Indicator friction error shall not exceed 2° C.

4.5.5 Insulation resistance. Insulation resistance shall be a minimum of 20 megohms.

4.5.6 Reduced voltage operation. The indicator shall be subjected to a Scale Error at Room Temperature test with an applied voltage of 22.5 V at the test points specified in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5 V applied by more than the tolerances specified in Table I.

4.5.7 Scale error at low temperature. Scale error shall not exceed the tolerances specified in Table I.

4.5.8 Scale error at high temperature. Scale error shall not exceed the tolerances specified in Table I.

4.5.9 Magnetic effect. The magnetic compass shall not deflect more than 5 degrees.

4.5.10 Resistance bulb current. A total resistance of 100 ohms shall be connected across pins "B" and "C" of the indicator connector. When the test voltage is applied across pins "A" and "C", the current in the resistance bulb circuit shall not exceed 22 ma.

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TABLE I
Test Tolerances

Test Point °C	Test Resistance Ohms	Scale Error at Room Temperature Tolerance ±°C	Reduced Voltage Tolerance ±°C	Scale Error at Low & High Temperature Tolerance ±°C	Scale Error at Room Temperature after Open and Short Circuit Tolerance ±°C	Scale Error at Room Temperature after Vibration Failure Tolerance ±°C
-50	74.24	1.5(1.5)	1.0(1.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
-30	80.56	1.5(1.5)	1.0(1.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
-10	87.04	1.5(1.5)	1.0(1.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
0	90.38	1.0(1.0)	1.0(1.0)	2.0(2.0)	1.5(1.5)	3.5(3.0)
+10	93.80	1.5(1.5)	1.0(1.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
+30	100.91	1.5(1.5)	2.0(2.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
+50	108.39	1.5(1.5)	2.0(2.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)

		Extreme Low Temperature Exposure			
Test Point °C	Test Resistance Ohms	Scale Error at Low Temperature Tolerance ±°C	Scale Error at Room Temperature after Low Temperature Exposure Tolerance ±°C	Scale Error at Room Temperature after High Temperature Exposure Tolerance ±°C	Scale Error with Bulb Lead Resistance Tolerance ±°C
-50	74.24	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
-30	80.56	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
-10	87.04	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
0	90.38	4.5(2.0)	3.5(1.0)	3.5(1.0)	1.5(1.5)
+10	93.80	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
+30	100.91	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
+50	108.39	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)

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4.5.11 Open circuit and short circuit. Scale error at room temperature after open and short circuit shall not exceed the tolerances specified in Table I.

4.5.12 Vibration error. When the indicator is subjected to the vibration error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 2° C. The maximum difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 2° C. Pointer oscillation shall not be considered as pointer variation.

4.5.13 Vibration failure. The scale error at room temperature after vibration shall not exceed the tolerances specified in Table I.

4.5.14 Response time. The response time shall not exceed 3 seconds.

4.5.15 Damping factor. The damping factor shall not be less than 1.6.

4.5.16 Power consumption. The power consumption shall not exceed 2.5 watts.

4.5.17 Low temperature exposure. Scale error at low temperature shall not exceed the tolerances specified in Table I. Scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I.

4.5.18 High temperature exposure. Scale error at room temperature after high temperature exposure shall not exceed the tolerances specified in Table I.

4.5.19 Bulb lead resistance effect. The indicator shall be connected as shown in Figure 3 and subjected to the Scale Error at room temperature test. The indications with bulb lead resistance shall not vary from the indications without bulb lead resistance by more than the tolerance specified in Table I.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Each indicator shall be preserved in accordance with Method II of MIL-P-116 and packaged in a container conforming to PPP-B-636, Class I, style optional unless otherwise specified by the procuring activity.

5.1.2 Level C. Each indicator shall be preserved in accordance with Method III of MIL-P-116 and packaged in accordance with Level C of MIL-P-7936.

5.2 Packing shall be to the level specified in the contract or order (See 6.3). The applicable level (A, B or C) shall be in accordance with MIL-P-7936.

5.3 Marking for shipment and storage. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

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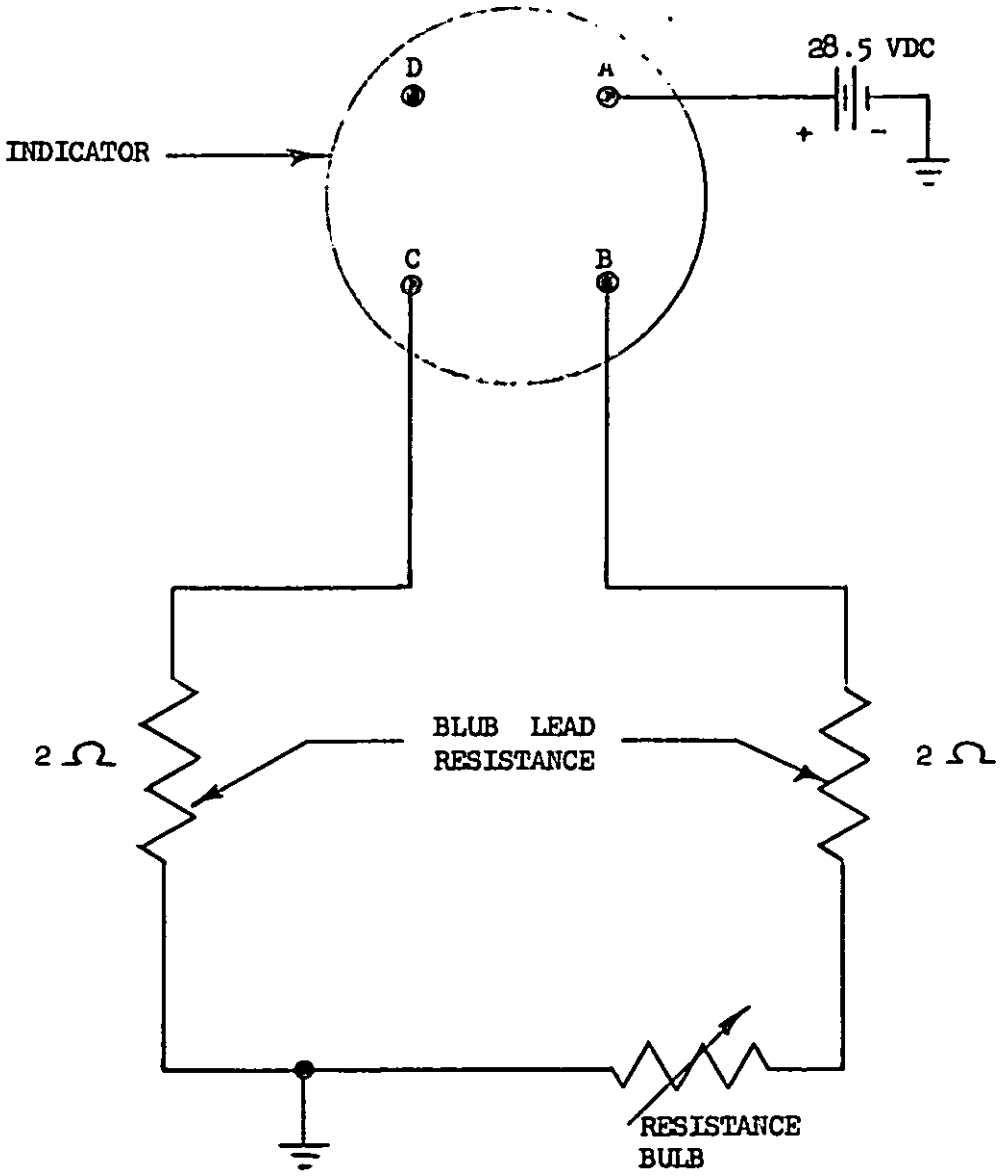


FIGURE 3. Wiring Diagram

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6. NOTES

6.1 Intended use. The indicator covered by this specification is intended to indicate temperature in the range of -50 to +50° C, such as free air, carburetor air, or cabin air.

6.2 General. The instructions contained in Section 6 of MIL-I-7070 are applicable to and form a part of this specification except that information pertaining to qualification of products covered by this specification shall be obtained from:

COMMANDER
MIDDLETOWN AIR MATERIEL AREA
ATTN: MANE
OLMSTED AIR FORCE BASE, PENNSYLVANIA 17057

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type, number of indicator desired.

c. Ordering data in accordance with MIL-I-7070.

Custodian:

Air Force - 67
Army - EL
Navy - WP

Preparing Activity:

Air Force - 67

Project Number 6685-0090

Reviewers:

Air Force - 67, 11
Army - MI, MO, GL
Navy - WP

Users:

Army - EL, SM

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: MIL-I-7075A Indicator, Temperature, Electrical Resistance, Multifunction, -50 to +50°C, 2 inch size

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

AF Olmsted Sep 63, 150

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